Spec → Code: RID013

Exported 8/20/2026, 1:35:46 PM · Tenant: default

1. Spec

REPORT ID

RID013

EDC SYSTEM / LANGUAGE

veeva / CQL

DESCRIPTION

This report is used to identify patients with last dose checkbox missing and missing/incorrect Label Description Status. (Report Title: Action Taken with Study Medication vs Study Medication Last Dose and Label Description Status reconciliation)

COLUMN HEADING

1. Country
2. Site
3. Subject Number
4. AE Seq
5. AE Term
6. AE Action Taken with Study Medication [X]
7. SM Visit
8. SM eCRF
9. SM Seq
10. Study Medication
11. Label Description/Dose Number
12. SM Status
13. Error Message

Form.Item Names:
1. SA.Country
2. SA.Site Mnemonic
3. SA.Subject Number
4. AE.FORMIDX
5. AE.AE Term
6. AE.Action taken with Study Medication [X]
7. SM/SMAU/SMAB.VISITMNEMONIC
8. SM/SMAU/SMAB.FORMMNEMONIC
9. SM/SMAU/SMAB.FORMIDX
10. SM/SMAU/SMAB.Study Medication
11. SM.Label Description/SMAU.Dose Number/SMAB.Dose Number
12. SM/SMAB/SMAU.Label description/Study Medication Status
13. Logic.Error Message

LOGIC

Logic 1
Logic 1.1 AE.Action Taken with Study Medication = Discontinued
AND
Logic 1.2 SM/SMAB/SMAU.Label Description/Study Medication = Study Medication
AND
Logic 1.3 SM/SMAB/SMAU.Label Description/Study Medication with maximum Stop Date/Time
AND
Logic 1.4 SM/SMAB/SMAU. Please Indicate if this is the Last Dose for this Label Description/Study Medication = null
Error Message: Last Dose Missing

Logic 2
Logic 2.1 AE.Action Taken with Study Medication = Discontinued

AND
{Logic 2.2 SM/SMAB/SMAU.Label Description/Study Medication = Study Medication
AND
Logic 2.3 SM/SMAB/SMAU.Please Indicate if this is the Last Dose for this Label Description/Study Medication <> null
AND
Logic 2.4 SM. SM/SMAB/SMAU.Label description/Study Medication Status = null}
Error Message: Label Description Status Missing

Logic 3
Logic 3.1 AE.Action Taken with Study Medication = Discontinued
AND
{Logic 3.2 SM/SMAB/SMAU.Label Description/Study Medication = Study Medication
AND
Logic 3.3 SM/SMAB/SMAU.Please Indicate if this is the Last Dose for this Label Description/Study Medication <> null
AND
Logic 3.4 SM. SM/SMAB/SMAU.Label description/Study Medication Status <> Adverse Event}
Error Message: Incorrect Label Description Status

Notes
1. For studies with multiple study medications, match AE action taken with Study Medication [x] and Relationship to Study Medication [x] with Study Medication [x] on SM/SMAU/SMAB. For studies with single study medication, Overall Action Taken and/or Overall Relationship to Study Medication is checked and a match on Study Medication name is not performed.

2. If pulldown lists Study Medication 1, Study Medication 2 etc. are available in the AE form, match the Action taken with Study Medication [x] and Relationship to Study Medication [x] with pulldown list value provided in Study Medication [x] and with Study Medication [x] on SM/SMAU/SMAB.

3. Exclude deleted data and erroneous visit from programming

7. For SM/SMAU/SMAB Study Medication, use SM.Label Description/SMAU.Dose Number/SMAB.Dose Number directly. If SM Label Description and SMAU/SMAB Dose Number are not exactly matched with Study Medication, using the optional supporting doc study_med_match_supp_.csv to define the mapping.

The optional supporting document is only required in the following cases:
- SM Label Description and SMAU/SMAB Dose Number are not exactly matched with Study Medication. This supporting document is only required to be created once per trial. If it was already created and uploaded, there is no need to duplicate it.

Optional supporting document should be named as study_med_match_supp_.csv and will contain the following columns and associated content:
1 row for each Label Description or Dose Number present in SM/SMAU/SMAB eCRFs.
1. eCRF – SM/SMAU/SMAB
2. Study Medication – Value of Study Medication
3. Matching Item – SM Label Description or SMAU/SMAB Dose Number
4. Matching Value – Value of SM Label Description or SMAU/SMAB Dose Number

eCRF,Study Medication,Matching Item,Matching Value
SM,-1439,Label Description,-1439 100 mg
SM,Methadone,Label Description,Methadone Tablet 5 mg
SM,Methadone,Label Description,Methadone Tablet 10 mg
SMAB,3475/Placebo,Dose Number,3475/Placebo
SMAU,Carboplatin,Dose Number,Carboplatin
SMAU,Carboplatin,Dose Number,Carboplatin - Dose 1

SMAU,Carboplatin,Dose Number,Carboplatin - Dose 2
SMAU,Carboplatin,Dose Number,Carboplatin - Dose 3

The actual information provided by the optional supporting doc can be outputted by selecting Y in the parameter "Study Med Match Page".

Parameter Selection: Country, First Site, Last Site, First Subject, Last Subject, List of Subjects, Randomized Only, Study Med Match Page (Yes/No)

REPORT SCENARIOS

Output: YES
1. Output will be displayed when Adverse Event action taken with Study Medication is marked as Discontinued, but Study Medication last dose checkbox is missing
2. Output will be displayed when Adverse Event action taken with Study Medication is marked as Discontinued, but Label Description Status for this Study Medication is missing.
3. Output will be displayed when Adverse Event action taken with Study Medication is marked as Discontinued, but Label Description Status for this Study Medication is different than Discontinued due to Adverse Event.

OUTPUT: NO
4. Deleted AE or SM/SMAB/SMAU entries.
5. Output will NOT be displayed when Adverse Event action taken with Study Medication is marked as Discontinued, and Label Description Status for this Study Medication is equal to Discontinued due to Adverse Event.
6. Output will NOT be displayed when Adverse Event action taken with Study Medication is marked with something different from Discontinued.

SORTING

Country, Site, Subject Number, Visit, AESeq, SM eCRF, SM Seq, Study Medication

2. Logic Understanding

This report cross-checks AE records where Action Taken with Study Medication = 'Discontinued' against corresponding SM/SMAU/SMAB study medication records (matched by Study Medication name/pulldown, possibly via a supplemental mapping file) to find three distinct problems: (1) the most recent (max Stop Date) dosing record has a missing 'last dose' checkbox, (2) an answered last-dose record has a missing Status field, or (3) an answered last-dose record has a Status that isn't the expected 'Discontinued due to Adverse Event'. Deleted records and erroneous visits are excluded, and each flagged row is output with its specific error message, sorted by Country/Site/Subject/Visit/Seq/Study Medication.

1. Start by identifying AE records where AE.Action Taken with Study Medication [X] = 'Discontinued'. This is a common precondition (Logic 1.1, 2.1, 3.1) across all three error branches - if the action taken is anything other than Discontinued, none of the checks apply (per Output:NO scenario 6).

2. For studies with multiple study medications, match the specific AE.Action Taken with Study Medication [X] (and Relationship to Study Medication [X]) to the corresponding SM/SMAU/SMAB record by Study Medication value, per Note 1; if the study has only a single study medication, use the Overall Action Taken/Relationship field instead and skip the Study Medication name match, per Note 1. *(inferred)*

3. If the AE form provides pulldown lists like 'Study Medication 1', 'Study Medication 2', use the pulldown value for Study Medication [X] to match against SM/SMAU/SMAB Study Medication, per Note 2.

4. When matching SM.Label Description or SMAU/SMAB.Dose Number to the Study Medication value is not an exact text match, consult the optional study_med_match_supp_.csv lookup file (per Note 7) to resolve the correct mapping between the eCRF's Label Description/Dose Number value and the Study Medication name.

5. Logic 1: Among the matched SM/SMAB/SMAU records for that Study Medication (Logic 1.2), identify the one record with the maximum Stop Date/Time (Logic 1.3) - i.e., the most recent dosing record for that medication.

6. Logic 1 continued: On that maximum-Stop-Date record, check if 'Please Indicate if this is the Last Dose for this Label Description/Study Medication' checkbox is null (Logic 1.4). If AE Action Taken = Discontinued AND the matched Study Medication's latest record has a null last-dose checkbox, flag error 'Last Dose Missing'.

7. Logic 2: Separately, for SM/SMAB/SMAU records matched to the Study Medication (Logic 2.2) where the last-dose checkbox is NOT null (Logic 2.3, i.e., it has been answered), check if the SM Label description/Study Medication Status field is null (Logic 2.4). If AE Action Taken = Discontinued AND last-dose checkbox is answered AND Status is null, flag error 'Label Description Status Missing'.

8. Logic 3: For the same set of records as Logic 2 (matched Study Medication, last-dose checkbox NOT null), check if the Status field is populated but NOT equal to 'Adverse Event' (i.e., not 'Discontinued due to Adverse Event') (Logic 3.4). If AE Action Taken = Discontinued AND last-dose checkbox is answered AND Status <> the expected adverse-event-related value, flag error 'Incorrect Label Description Status'.

9. Logic 2 and Logic 3 appear mutually exclusive in coverage of the Status field: Logic 2 catches Status = null, Logic 3 catches Status populated but incorrect; together they fully partition the 'checkbox answered' population between missing-status and wrong-status errors, while Status = 'Discontinued due to Adverse Event' (the correct value) yields no error, per Output:NO scenario 5. *(inferred)*

10. Before applying any of the above logic, exclude deleted AE records, deleted SM/SMAB/SMAU entries, and records tied to erroneous visits, per Note 3 and Output:NO scenario 4.

11. For each flagged record (one of the three error conditions), output the identifying columns (Country, Site, Subject Number, AE Seq, AE Term, AE Action Taken, SM Visit, SM eCRF, SM Seq, Study Medication, Label Description/Dose Number, SM Status) along with the corresponding Error Message ('Last Dose Missing', 'Label Description Status Missing', or 'Incorrect Label Description Status').

12. Optionally, if the parameter 'Study Med Match Page' is set to Yes, output an additional page/section showing the content of the study_med_match_supp_.csv mapping file used for the Study Medication matching.

13. Sort final output by Country, Site, Subject Number, Visit, AE Seq, SM eCRF, SM Seq, Study Medication as specified in the Sorting field.

3. Spec Review

Readiness: Needs revision — The spec has a solid overall structure but contains a blocking ambiguity in how Logic 2 and Logic 3 distinguish null vs. non-'Adverse Event' status values (risking duplicate/overlapping output rows), plus a wording mismatch between Logic 3.4 and its corresponding Report Scenario/5 text. Several other warnings around the single-medication matching path and undefined tie-breaking for maximum Stop Date/Time should be resolved before handing off to a query generator.

BLOCKING (consistency): Report Scenario 3 says output occurs when 'Label Description Status for this Study Medication is different than Discontinued due to Adverse Event,' but Logic 3.4 states the condition as SM/SMAB/SMAU Status <> 'Adverse Event' (not '<> Discontinued due to Adverse Event'). It's unclear whether 'Adverse Event' in Logic 3.4 is shorthand for the full status value 'Discontinued due to Adverse Event' referenced in the scenario and in Scenario 5, or a distinct, differently-named status option. → Confirm the exact allowed value list for SM/SMAB/SMAU Status and use the identical literal value in Logic 3.4 and Scenario 3/5 text (e.g., both should read 'Discontinued due to Adverse Event').

BLOCKING (testability): Logic 2 and Logic 3 share identical trigger conditions (2.1/2.2/2.3 vs 3.1/3.2/3.3 are the same: Action Taken = Discontinued, Study Medication match, Last Dose indicator <> null) and differ only on the status field being null (Logic 2.4) vs <> 'Adverse Event' (Logic 3.4). Since 'null' and '<> Adverse Event' are not stated as mutually exclusive in the spec (a null value could arguably also satisfy '<> Adverse Event' depending on how the CQL/database represents null comparisons), it's ambiguous whether a record with a null status would trigger Logic 2, Logic 3, or both, producing two output rows for the same discrepancy. → Explicitly state that Logic 3.4 excludes null values (e.g., 'Status <> Adverse Event AND Status is not null') so Logic 2 and Logic 3 are mutually exclusive, or clarify that null values are only ever captured by Logic 2.

WARNING (ambiguity): Logic 1.3 states 'SM/SMAB/SMAU.Label Description/Study Medication with maximum Stop Date/Time' but does not specify what happens when there are multiple records tied for maximum Stop Date/Time, or how this interacts with records where Stop Date/Time is null (e.g., ongoing medication with no stop date). → Add explicit tie-breaking logic for multiple records sharing the same maximum Stop Date/Time, and state how records with null/missing Stop Date/Time (ongoing medications) should be treated for this maximum determination.

WARNING (completeness): Note 1 describes matching AE Action Taken/Relationship to Study Medication with SM/SMAU/SMAB by Study Medication[x] name for multi-medication studies, and a different path (Overall Action Taken, no name match) for single-medication studies, but the Logic section (1.1-1.4, 2.1-2.4, 3.1-3.4) never references 'Overall Action Taken' or shows how the single-medication path is implemented in the actual logic conditions. → Add explicit logic branches (or a clear statement) showing how Logic 1-3 differ or are applied when the study has a single study medication and Overall Action Taken is used instead of Action Taken with Study Medication [X].

WARNING (ambiguity): Note 2 references matching 'Relationship to Study Medication [x]' as part of the AE/SM matching logic, but this field never appears anywhere in the Logic conditions (Logic 1-3 only use Action Taken with Study Medication) or in the Column Headings/Form.Item list. → Either remove the Relationship to Study Medication reference from Note 2 if it isn't used in this report's logic, or add it explicitly to the Logic conditions and Form.Item list if it is required for matching.

INFO (completeness): Note 3 says to 'Exclude deleted data and erroneous visit from programming' and Report Scenario 4 says output will not display 'Deleted AE or SM/SMAB/SMAU entries,' but neither the Logic section nor the Form.Item list references an erroneous-visit flag/field to implement this exclusion. → Add the specific item name(s) used to identify deleted records and erroneous visits (e.g., a status/flag field) to the Form.Item Names list so the exclusion can be implemented consistently.

INFO (naming): Form.Item Name 12 is written as 'SM/SMAB/SMAU.Label description/Study Medication Status' while Column Heading 12 is 'SM Status' and Logic 2.4/3.4 refer to it as 'SM. SM/SMAB/SMAU.Label description/Study Medication Status' (with an apparent stray 'SM.' prefix before the eCRF list). The naming is inconsistent in format across these three locations. → Standardize the item reference to a single consistent form, e.g., 'SM/SMAB/SMAU.Label Description/Study Medication Status', and remove the stray leading 'SM.' in Logic 2.4 and 3.4.

INFO (ambiguity): Item 11 is named 'SM.Label Description/SMAU.Dose Number/SMAB.Dose Number' and the supporting doc mapping in Note 7 discusses matching this against 'Study Medication' when not an exact match, but the spec doesn't clarify whether the Column Heading 'Label Description/Dose Number' in the final output reflects the raw eCRF value or the mapped 'Study Medication' value from the supporting doc. → Clarify whether Column Heading

> 11 displays the raw SM/SMAU/SMAB captured value or the mapped Study Medication value when the supporting document is used.

4. Generated Code

CQL — Guardrails: passed

```
SELECT ae.`@HDR`.`Site`.`Country` AS `Country`, ae.`@HDR`.`Site`.`Number` AS `Site`,
ae.`@HDR`.`Subject`.`Name` AS `Subject Number`, ae.AESPIDIT AS `AE Seq`, ae.AEAETERMIT AS `AE
Term`, ae.AEAEACNSMIT AS `AE Action Taken with Study Medication [X]`, sm.`@HDR`.`Event`.`Name` AS
`SM Visit`, sm.`@Form`.`Name` AS `SM eCRF`, sm.SMEXSPIDIT AS `SM Seq`, sm.SMEXTRTIT AS `Study
Medication`, sm.SM_LBL1 AS `Label Description/Dose Number`, sm.SMEXPRDFLAGIT AS `SM Status`, 'Last
Dose Missing' AS `Error Message`
FROM `EDC`.`AEFM` ae
JOIN `EDC`.`SMFM` sm ON ae.`@HDR`.`Subject`.`ID` = sm.`@HDR`.`Subject`.`ID` AND sm.SMEXTRTIT =
ae.AEAESMIT
JOIN (SELECT sm2.`@HDR`.`Subject`.`ID` AS Subject_ID, sm2.SMEXTRTIT AS StudyMed,
MAX(RAWDATE(sm2.SMEXENDTCIT)) AS MaxStopDate
FROM `EDC`.`SMFM` sm2
WHERE sm2.`@Form`.`Status` = 'submitted__v' AND (sm2.DELETEDFM = 'N' OR sm2.DELETEDFM IS NULL)
GROUP BY sm2.`@HDR`.`Subject`.`ID`, sm2.SMEXTRTIT) maxsm ON maxsm.Subject_ID =
sm.`@HDR`.`Subject`.`ID` AND maxsm.StudyMed = sm.SMEXTRTIT AND RAWDATE(sm.SMEXENDTCIT) =
maxsm.MaxStopDate
WHERE ae.`@Form`.`Status` = 'submitted__v' AND sm.`@Form`.`Status` = 'submitted__v' AND
(ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL) AND (sm.DELETEDFM = 'N' OR sm.DELETEDFM IS NULL) AND
UPPER(ae.AEAEACNSMIT) = 'DISCONTINUED' AND sm.SMEXLDINDCCB IS NULL
UNION ALL
SELECT ae.`@HDR`.`Site`.`Country` AS `Country`, ae.`@HDR`.`Site`.`Number` AS `Site`,
ae.`@HDR`.`Subject`.`Name` AS `Subject Number`, ae.AESPIDIT AS `AE Seq`, ae.AEAETERMIT AS `AE
Term`, ae.AEAEACNSMIT AS `AE Action Taken with Study Medication [X]`, sm.`@HDR`.`Event`.`Name` AS
`SM Visit`, sm.`@Form`.`Name` AS `SM eCRF`, sm.SMEXSPIDIT AS `SM Seq`, sm.SMEXTRTIT AS `Study
Medication`, sm.SM_LBL1 AS `Label Description/Dose Number`, sm.SMEXPRDFLAGIT AS `SM Status`, 'Label
Description Status Missing' AS `Error Message`
FROM `EDC`.`AEFM` ae
JOIN `EDC`.`SMFM` sm ON ae.`@HDR`.`Subject`.`ID` = sm.`@HDR`.`Subject`.`ID` AND sm.SMEXTRTIT =
ae.AEAESMIT
WHERE ae.`@Form`.`Status` = 'submitted__v' AND sm.`@Form`.`Status` = 'submitted__v' AND
(ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL) AND (sm.DELETEDFM = 'N' OR sm.DELETEDFM IS NULL) AND
UPPER(ae.AEAEACNSMIT) = 'DISCONTINUED' AND sm.SMEXLDINDCCB IS NOT NULL AND sm.SMEXPRDFLAGIT IS NULL
UNION ALL
SELECT ae.`@HDR`.`Site`.`Country` AS `Country`, ae.`@HDR`.`Site`.`Number` AS `Site`,
ae.`@HDR`.`Subject`.`Name` AS `Subject Number`, ae.AESPIDIT AS `AE Seq`, ae.AEAETERMIT AS `AE
Term`, ae.AEAEACNSMIT AS `AE Action Taken with Study Medication [X]`, sm.`@HDR`.`Event`.`Name` AS
`SM Visit`, sm.`@Form`.`Name` AS `SM eCRF`, sm.SMEXSPIDIT AS `SM Seq`, sm.SMEXTRTIT AS `Study
Medication`, sm.SM_LBL1 AS `Label Description/Dose Number`, sm.SMEXPRDFLAGIT AS `SM Status`,
'Incorrect Label Description Status' AS `Error Message`
FROM `EDC`.`AEFM` ae
JOIN `EDC`.`SMFM` sm ON ae.`@HDR`.`Subject`.`ID` = sm.`@HDR`.`Subject`.`ID` AND sm.SMEXTRTIT =
ae.AEAESMIT
WHERE ae.`@Form`.`Status` = 'submitted__v' AND sm.`@Form`.`Status` = 'submitted__v' AND
(ae.DELETEDFM = 'N' OR ae.DELETEDFM IS NULL) AND (sm.DELETEDFM = 'N' OR sm.DELETEDFM IS NULL) AND
UPPER(ae.AEAEACNSMIT) = 'DISCONTINUED' AND sm.SMEXLDINDCCB IS NOT NULL AND sm.SMEXPRDFLAGIT IS NOT
NULL AND UPPER(sm.SMEXPRDFLAGIT) <> 'DISCONTINUED DUE TO ADVERSE EVENT'
ORDER BY `Country`, `Site`, `Subject Number`, `SM Visit`, `AE Seq`, `SM eCRF`, `SM Seq`, `Study
Medication`
```

Only SMFM exists in the confirmed study metadata (no SMAU/SMAB forms were found), so all SM/SMAU/SMAB logic in the spec is implemented against SMFM only -- this should be verified, since the spec assumes multiple SM-family forms may exist. Country/Site/Subject Number are pulled from @HDR per platform convention rather than the SA form referenced in the spec's Form.Item Names. AE Seq/AE Term are mapped to AESPIDIT (matching the confirmed pattern from a prior verified AE report) and AEAETERMIT (the only term item present for AEFM in this study, since no bare AETERMIT exists); the Action Taken item is assumed to be AEAEACNSMIT and the per-medication AE selector used for matching to SM.SMEXTRTIT is assumed to be AEAESMIT -- both should be confirmed via DESCRIBE FORM. The "Last Dose" checkbox confidently maps to SMEXLDINDCCB, but no item in SMFM's known metadata is clearly named "Label Description" or "Study Medication Status"; SM_LBL1 and SMEXPRDFLAGIT were used as closest available placeholders and are flagged as unverified guesses that must be validated before use. Deleted records are excluded via the tolerant-null DELETEDFM pattern on both forms;

"erroneous visit" exclusion and the optional Study Med Match Page CSV output could not be mapped to any known Form/Item and are therefore not implemented (the CSV is external reference data, not Form-sourced), and runtime parameters (Country/Site/Subject ranges, Randomized Only) are assumed to be applied by the reporting tool rather than embedded in this CQL since the spec's Logic section does not define their filtering behavior.